RESEARCH AND PROCUREMENT PLANNING

Agricultural Spray Drones

American Manufacturers, Costs, and a Cuesta Training Path

Evidence reviewed October 15, 2026

A decision aid for curriculum and equipment planning. Listed prices are not binding quotations; manufacturer claims, delivery evidence, and regulatory approvals are distinguished throughout. PEGASUS delivery remains unverified despite a published package price.

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Executive assessment

Hylio is a strong starting point for a Cuesta College shortlist when the goal is an American-built agricultural sprayer with an established product line, a published training ecosystem, and reasonably transparent equipment pricing. The compact PEGASUS merits a teaching-focused evaluation; the larger ARES merits a commercial-application evaluation. **PEGASUS is still advertised for preorder/ coming soon by a reviewed dealer, and immediate delivery was not verified.** ARES has the stronger current production-platform evidence. Those are different purchases. A larger tank increases productivity, but it also increases aircraft energy, ground-support requirements, regulatory work, and the consequences of student error.

Central UAS Technologies, formerly Leading Edge, should also receive a serious quote request: its PrecisionVision family has documented university agricultural-research use. The broader American market needs careful classification. Ceres Air markets a domestically assembled platform with global components. Revolution has delivered Brazilian-origin I-19 units while developing U.S. manufacturing. EXEDY offers a U.S.-manufacturing proposition under a Japanese parent. Elevon and Agricision deserve attention as newer entrants, with launch, delivery, and support maturity still requiring verification. Pyka is a credible American manufacturer in a much larger industrial aircraft and budget class. Rotor/ Robinson's Sprayhawk belongs in that industrial category too, with a material U.S. approval caveat. Guardian's SC1 should not be placed on a current procurement shortlist based on old articles.

The price gap is meaningful, but package definitions matter more than headline airframe prices. A dated, delivery-unverified PEGASUS configuration is $28,444; ARES dealer packages are approximately $46,000. Some imported packages are substantially less expensive. These are equipment prices, not complete college-program budgets. The worked scenarios below put a compact domestic teaching setup near $58,000 and a more capable domestic mobile setup near $88,000 in first-year planning dollars, including explicit allowances. They are estimates to refine through quotations, not spending approvals or offers from vendors.

For Cuesta, the recommended sequence is to define the instructional activity and qualified supervision, obtain comparable written quotes for Hylio and Central UAS, and evaluate emerging domestic alternatives against the same acceptance test. Keep one import quote as a cost and workflow benchmark if district and funding rules permit it. A partner-led demonstration can establish curriculum value before ownership, especially while the college resolves the California pesticide-pilot pathway. No vendor has been contacted and no purchase is recommended without those gates.

Scope and how to read the evidence

This is a broad U.S.-market survey of manufacturers and materially different aircraft families with public evidence of agricultural liquid dispensing. It includes conventional multirotors, industrial fixed-wing aircraft, unmanned helicopters, emerging U.S. assembly programs, and a small set of custom integrations. It deliberately separates crop imaging, granular spreading, spraying services, and heavy-lift platforms from complete agricultural sprayers. It is comprehensive within the accessible public evidence reviewed, not a claim that every global OEM, white-label seller, or private prototype has been found.

Manufacturer pages establish what a maker currently claims. Dealer pages establish a listed asking price and advertised package, not a binding quote, guaranteed stock, or independent proof of reliability. Regulatory sources establish rules and published approval status, not Cuesta's permission to operate a particular configuration. Recommendations and budget allowances are the analysis in this report. All prices are U.S. dollars unless explicitly stated, observed October 15, 2026; taxes, freight, duties, training travel, accessories, and recurring services are included only when identified. Older dated price sheets are labeled as such. The research did not involve checkout, serial-number stock confirmation, a factory audit, customer-reference calls, or flight testing.

The repository's July program discussion places agriculture alongside development of the basic and advanced-mapping courses. This report supports a separate, phased spraying decision without

assuming that an earlier general equipment budget remains available. It does not revise course promises, existing purchasing decisions, or the college's operating permissions. Program context

What "American" should mean in this procurement

Use separate fields for corporate ownership, design responsibility, final assembly, manufacturing of critical parts, software and data control, and the exact legal purchasing test. A U.S. headquarters does not establish American ownership of every parent company. American assembly does not establish that motors, flight controllers, radios, sensors, battery cells, or software are American. A domestic company using imported components may still be a useful option; the disclosure should be explicit.

The FTC's unqualified Made in USA standard is materially stronger than a statement that final assembly happens here. NDAA representations, Green UAS certification, Blue UAS listing, FCC equipment authorization, FAA operating approval, and Buy American eligibility answer different questions. Require a model-specific evidence package for the delivered configuration and the actual source of college funding. The regulatory section explains these distinctions and the October 2026 changes. FTC origin guidance

American-options comparison at a glance

These are advertised equipment prices, not delivered system quotations. The following manufacturer profiles provide the supporting links, package details and origin evidence. "Quote required" means no dependable public current price was established, not that the aircraft is necessarily unavailable. Availability statements are observations as of October 15, 2026.

Candidate and liquid capacity	Origin evidence	Public equipment price	Maturity and Cuesta fit
Hylio PEGASUS, 2.5 gal	Texas design, software and assembly; global parts possible	$28,444 dated three-battery/ charger package	**Preorder/coming soon; delivery unverified.** Promising small trainer; weigh configured aircraft below 55 lb if using that route.
Hylio ARES, 13 gal	Texas-built with U.S. and global parts; independently listed Green UAS	$45,999 named dealer kit; $47,449 Titan estimate	Priority production-oriented domestic shortlist; heavy-aircraft operating pathway.
Hylio ATLAS, 30 gal	Hylio domestic platform, exact configuration evidence required	$64,998 advertised base, not complete field system	Sold-out listing and 2026 delivery forecast; verify actual deliveries. Large production/research use.
Ceres Black Betty, 31/40 gal options	U.S. assembly/testing; over 30% U.S. cost content claimed, higher targets are future	$57,998 six-battery kit; exact configuration unresolved	Current marketed platform; stock, power and school-account terms need written confirmation.
Revolution I-19, 19 gal official load	Initial units imported from Brazil; U.S. production developing	$37,950 one-battery kit; $46,250 three-battery/RTK kit	Deliveries documented; large-aircraft comparator, not wholly American-made. Other models have different schedules.
Elevon Z30 / Z50 / Z80, about 8 / 13 / 21 gal	EFT-derived airframes, U.S. assembly and adaptations, international components	$23,950 / $28,950 / $36,950 advertised; power-kit scope unclear	Fall-2026 launch. Delivery and long-term support need validation; none is sub-55 lb loaded.
EXEDY AYRE CX, 15 gal	Japanese parent; Turkish technology lineage; U.S. manufacturing claimed	Quote required	September-2026 production plan, delivery unverified. Emerging quote/demo candidate.
Agricision BUBBA-18G, 70 L	Virginia design/manufacture and allied components claimed	Quote required	Manufacturer still says testing. Research watchlist, not a confirmed immediate purchase.
Central UAS PrecisionVision 100X, 50 L / 100 lb limit	Florida design and assembly; domestic and global components	Current quote required; historical 40X estimate is not a 100X price	Established application lineage and university research; confirm current model, delivery and support.
Pyka Pelican 2, 80 gal	U.S. manufacturing and Alameda facility; component origins not fully established	$550,000 to Nov. 14; $580,000 from Nov. 15, 2026	Commercial operating evidence, industrial fixed-wing system. Runway and professional aviation partnership needed.
Rotor/Robinson Sprayhawk, 100/110 gal conflict	U.S. helicopter/autonomy businesses; exact supply chain unverified	Quote required; old promotional price expired	Current OEM page says U.S. spray approval pending. Helicopter-scale research/ operator pathway.
Aero Systems West + LahakX	U.S.-manufactured carrier claim; ASW has Japanese parent	Custom integration/quote	Relevant California partnership lead; no complete priced retail spraying package verified.
Guardian SC1	Historical U.S. developer	No current supported new-aircraft price	Completed 2025 asset auction and no verified new-sales channel. Exclude from current purchasing shortlist.

American manufacturers and emerging domestic programs

Hylio: well-documented US-designed multirotors, with model-specific availability

Hylio describes proprietary hardware design and software development in-house, US suppliers for various structural materials, some internally manufactured electronic and mechanical components, and final assembly and flight testing in Texas. This is substantial US design and manufacturing evidence, but it is not evidence that every motor, cell, semiconductor, radio, or raw material is American. Its description explicitly allows externally supplied components. Hylio manufacturing process

The current ARES HYL-150 has a 13-gallon/50-liter liquid tank, an optional 20-gallon/76-liter granular hopper, and a 110-pound payload limit. Hylio advertises up to 70 acres/hour at 2 gallons/acre for spraying and 120 acres/hour at 20 pounds/acre for spreading. These are separate configurations and application rates. The published takeoff weight is 220 pounds, with an 80-pound airframe excluding battery. Hydraulic nozzles and rotary atomizers are alternatives. The manufacturer states that ARES is built in Richmond, Texas using US and global parts. ARES product specifications

AUVSI independently lists the HYL150 ARES among completed Green UAS certifications. That is materially stronger evidence than a dealer merely writing “NDAA compliant.” The certification framework addresses cybersecurity and supply-chain concerns. It does not establish an entirely domestic bill of materials, and the ARES listing must not be extended automatically to PEGASUS, ATLAS, or legacy Hylio aircraft. The live Blue UAS roster could not be retrieved in this research pass, so independent Blue status should be checked separately rather than inferred from sales copy. AUVSI cleared list

Titan Pro’s January 2026 ARES sheet shows an estimated recommended package of $47,449. It comprises the aircraft, GroundLink controller, FPV camera, obstacle sensors, toolkit, lifetime AgroSol access, spray equipment, spare-parts kit, three batteries, and one 9 kW charger. Battery pricing is $2,550 each and charger pricing $2,750. The optional RTK base costs $1,500 and the dry spreader $1,699; DigiFarm corrections cost $850/year per GroundLink. These are outside the displayed package. Its line-item layout should be checked visually before allocating the separate spray and spare-kit subtotals. Prices are expressly subject to change. Titan ARES package

Another dealer advertises an ARES base price of $45,999 but labels its listing sold out. Its customizable scope and unspecified battery counts make that a weaker budget anchor than a named inventory unit with an explicit equipment list. It is evidence of an advertised price, not stock. Drone Spray Pro ARES listing Apple Farm Service inventory #77390 separately advertises $45,999 including three batteries and a 9 kW charger; retain that inventory-specific equipment list when comparing its price. Apple inventory

Hylio includes simulator access, videos, manuals, checklists, regulatory application guidance, and a one-year manufacturer warranty. It offers demonstrations and describes repair training and a dealer support network. Those benefits do not imply that unlimited hands-on instruction for successive student cohorts is free. Hylio ordering and support Titan Pro explicitly prices its two-day group TakeOff training at $1,500 and one-on-one SkyHigh training at $2,500. Its policy requires another package for later staff members, limits the group format to four organizations with three people each, and lists labor at $100/hour for customers who purchased there versus $200/hour otherwise. Parts and shipping are additional. This matters for instructor turnover and annual college intakes. Titan training and service

PEGASUS HYL-030 is the particularly interesting teaching-scale prospect. Hylio’s specification gives 2.5 gallons liquid, 4 gallons granular volume, and a 21-pound material payload in either configuration near sea level. Maximum recommended takeoff weight is 54.99 pounds. The bare aircraft is 13 pounds, battery 10 pounds, hydraulic application equipment 10 pounds, and rotary or granular equipment 11 pounds. Those rounded figures sum to 55 pounds with the full 21-pound payload in the latter

configurations. Therefore a college must weigh the actual configured aircraft, including accessories and contents, rather than infer legal weight from tank volume or rounded brochure figures. Four gallons describes dry-hopper volume, not permission to carry four gallons of water. The sheet also requires payload reduction with density altitude. Spray productivity is up to 13 acres/hour at 2 GPA. Hylio PEGASUS specification hosted by Titan

Titan's 23 February 2026 PEGASUS package estimate is $28,444. It includes the drone, controller, FPV, sensors, toolkit, lifetime AgroSol, spray equipment, spares, three 16 Ah batteries, and one 9 kW charger. Optional extras are the 4-gallon spreader at $1,499, RTK base at $1,500, DigiFarm at $850/year, and range-extension kit at $1,135. Batteries are $899 each. The price sheet's headline pump rate conflicts with its detailed no-nozzle flow figure, so use the separate specification sheet for nozzle-dependent application planning. Titan PEGASUS package

Availability remains qualified. MultiCopter Warehouse calls PEGASUS preorder/coming soon. Hylio's SEC-hosted offering memorandum says initial PEGASUS, ATLAS, and PHOTON deliveries were planned for 2026 and labels these forward-looking statements. Neither demonstrates that a particular college can receive a PEGASUS immediately. Dealer preorder, Hylio offering memorandum

ATLAS HYL-300 is advertised with 30 gallons liquid, a 50-gallon dry hopper, a 250-pound payload, and up to 150 acres/hour at 2 GPA. Drone Spray Pro lists $64,998 base and sold-out status. Its standard description includes aircraft, selected application equipment, FPV, radar, controller/software and support materials; power, RTK, and spares vary by configuration. Two batteries are used per flight, so a vaguely described base package is especially unsuitable for turnkey budgeting. This is a large-field or production-research candidate, not the default introductory trainer. ATLAS dealer specifications and price

The legacy AG-210, AG-216, AG-230 and AG-272 remain relevant to used equipment and dealer inventory. An older Frontier sheet records 2.6-, 4.2-, 8-, and 18-gallon liquid tanks respectively. Do not treat those historical specifications as current production or use them interchangeably with the new HYL series. The 210's small-aircraft appeal needs an exact configured-weight check just as PEGASUS does. Legacy lineup sheet

Central UAS Technologies: established application lineage and university research

Central UAS Technologies, formerly Leading Edge Aerial Technologies, belongs in the domestic shortlist. Its PrecisionVision 100X manufacturer sheet specifies a 50-liter liquid tank, separate 70-liter granular system, and 100-pound material payload limit, with Florida design, engineering and assembly using domestic and global components. The weight limit still applies when the nominal tank volume could hold more material. The maker offers nozzle-boom customization and proprietary flight/data tools; those may suit deposition and application-research labs, subject to a demonstration and software terms. Neither domestic assembly nor its Section 889 representation establishes independently verified Blue/Green certification. 100X manufacturer sheet Application history

Earlier models include the 25-pound-payload 35X and 40-pound-payload 40X. The manufacturer's indexed 40X page labels it legacy, no longer produced, with continued support. Several main-site URLs redirected or failed during this review, so confirm the currently supplied generation and service contact rather than assume every historical model remains orderable. Manufacturer's FAA-filed 35X technical description Indexed manufacturer 40X page

Mississippi State documents agricultural research using the 40X, which is meaningful institutional-use evidence. Its separate proposed sole-source purchase notice 234-114 estimated $62,710 for a 40X purchase and $18,400 for refurbishment of an existing PrecisionVision
22. The combined $81,110 is not a single new-aircraft price. The notice
is historical, has no visible issue date in its text, and does not itemize a comparable battery/charger package. It is not a current 100X quote or evidence that Cuesta should use sole-source procurement. No dependable current retail package price or training entitlement was established; request an itemized current proposal. MSU research account MSU institutional cost example

Pyka: genuine American autonomous spray aircraft, fundamentally different infrastructure

Pyka describes Pelican 2 as commercially operating with customers today. It is a fixed-wing electric agricultural aircraft with a 300-liter/80-gallon spray capacity, 300 kg/660-pound payload, and 635 kg/1,400-pound maximum takeoff weight. It advertises up to 260 acres/hour at 10 L/ha, approximately 1.07 GPA, or 178 acres/hour at 20 L/ha, approximately 2.14 GPA. Its wingspan is 38 feet and recommended operating runway 820 by 26 feet. The currently displayed ready-to-fly price is $550,000 through 14 November 2026 and $580,000 from 15 November. The package explicitly includes five hot-swap battery sets, fifteen packs total, a charging cart capable of charging four sets together, and two weeks of pilot training. Pelican 2 specification and dated pricing

Pyka's February 2025 launch announcement cites commercial predecessor operations beginning in 2021, deployments with Dole in Honduras and SLC Agrícola in Brazil, a US delivery to Heinen Brothers, and a twenty-aircraft Brazilian Pelican 2 order. Orders, earlier-model deliveries, and current-model installations should remain distinct. This is materially more evidence of agricultural deployment than an unflown concept, but institutional buyers still need the exact Pelican 2 delivery slot, US operating authorization, and support scope in their proposal. Pelican 2 launch and operating history

Pyka describes US manufacturing and an Alameda production facility. That supports US manufacture, not an all-US component-origin finding. Pyka manufacturer site For a college, this is an agricultural-aviation research partnership with runway access, aircraft handling, electrical infrastructure, insurance, and professional operational oversight. It is not an interchangeable alternative to a small quadcopter; the low advertised cost per acre does not eliminate the six-figure capital commitment or its distinct operational burden. These suitability judgments are analysis rather than manufacturer performance promises.

Rotor Technologies and Robinson Unmanned: heavy helicopter platform, US spray approval still qualified

Robinson's current R44 Sprayhawk page describes a gasoline-powered unmanned helicopter with a 2,500-pound maximum takeoff weight, two-person pilot/visual-observer crew, VLOS operation, and advertised 200-plus acres/hour at 2-plus GPA on large fields. Its narrative says 110 gallons while its specifications say 100 gallons. Obtain the controlled configuration sheet rather than silently choosing a number. Most importantly, that same current OEM page marks US Section 44807 approval as pending while describing approvals in other jurisdictions. It lists the spraying system, flow control, AGNAV terminal and support equipment, but does not provide a public turnkey price or unambiguous inclusion list. Robinson R44 Sprayhawk

Robinson's March 2026 announcement establishes a Robinson Unmanned business unit and a strategic agreement with Rotor for RPX autonomy. It identifies Ascent AeroSystems as wholly owned; it does not say Robinson acquired Rotor. Keep the corporate relationship precise. Robinson Unmanned announcement

Rotor's November 2024 demonstration announcement reported two completed prototypes and expected 2025 deliveries. Its early-adopter pricing expired in December 2024. Historical sub-$1 million figures should therefore not be represented as a live quote. Rotor Texas demonstration The current manufacturer describes New Hampshire development, but the aircraft's component origins and procurement eligibility need configuration-specific verification. This is appropriate only for a college already equipped for helicopter-scale aviation research or an operator partnership, with a written US authorization and delivery plan. Rotor manufacturer site

Ceres Air: US assembly with disclosed global sourcing, large-volume system

Ceres explicitly frames American manufacture as a progression: current US cost content exceeds 30%, with US assembly/testing and US-hosted software, while higher localization percentages are goals. Those disclosures are more useful than treating its branding as proof of a completely American bill of

materials. Its site advertises 31- and 40-gallon tank options. Ceres manufacturer and localization statement The Black Betty Stacked page describes 40 gallons, 11 GPM and a 397-pound lift demonstration. That lift figure is not automatically an authorized liquid payload or routine maximum takeoff weight. Black Betty Stacked

The manufacturer store displays $57,998 for a C31 Black Betty Stacked six-battery kit. However, the rendered product page also reports that the selected combination does not exist, and does not fully enumerate the charging and control package. Treat this as a publicly advertised kit price, not proven stock. Ceres six-battery kit A dealer listing expands a similarly priced package to drone, remote, six smart batteries and two chargers, but its authorization was not independently established in this pass. Obtain the actual manufacturer's or verified dealer's written bill of materials. Red Solutions listing

Ceres documentation describes training and approved pilot-account requirements. Those deserve special scrutiny for student account turnover and whether instructors can activate new cohorts without delay. Ceres knowledge base Its maintenance manual calls for monthly minor and quarterly major maintenance, with authorized service involvement. C31 maintenance manual Warranty terms vary by component, rather than promising one uniform term for the whole system. C31 warranty Charger descriptions differ between public pages, including a manufacturer support reference to three-phase 380 V, so electrical requirements must be specified before procurement. Ceres support specifications

The evidence supports a real, marketed platform with demonstrations and documentation. It does not establish a guaranteed delivery date, audited domestic content, or a campus-ready service commitment. Large tanks are useful for production work but make this a less natural first instructional aircraft than a verified small, weight-limited platform.

Elevon: transparently US-assembled new entrant, launch status needs confirmation

Elevon says it was formed in 2025 and uses EFT airframes modified with US software and hardware. Its about page describes the 2026 Z-series launch for fall 2026 and assembly in the US using domestic and international components, with increased American manufacture a goal. This supports US assembly and adaptation, not exclusive US design or components. The launch timing overlaps this research date; listing products and prices is insufficient proof of deliveries. Elevon company and sourcing

The Z30 is advertised at $23,950 with an approximately 8-gallon spray capacity, 65-pound granular capacity, 54-pound empty weight and 146-pound maximum takeoff weight. Its claimed 30 acres/hour lacks an accompanying application rate. The page explicitly identifies universities and test plots as uses, but the aircraft is not a sub-55-pound system when operationally loaded. Z30

The Z50 is advertised at $28,950, with 13 gallons liquid, a 100-pound granular capacity, 81-pound empty weight and 224-pound maximum takeoff weight. Claimed productivity is 40 acres/hour without a stated application rate. Z50 The Z80 is advertised at $36,950, with 21 gallons liquid, 172-pound granular capacity, 120-pound empty weight and 330-pound maximum takeoff weight. Its manufacturer calls for two-person handling and claims more than 50 acres/hour without a specified GPA. Z80

Those pages describe registration and regulatory-document support, but do not provide sufficiently exact battery/charger counts to treat prices as complete field packages. Training, warranty, local repair arrangements, power supply, additional batteries, freight and timing need a quote. An FCC conditional approval is not interchangeable with Green/Blue certification or proof of all-American sourcing. Avoid copying expiry dates from marketing without checking the latest FCC notice. For a college, Elevon is a credible demonstration-and-quote candidate, with acceptance testing and delivery protections particularly important because of its young product line.

Revolution: delivered I-19, evolving US manufacturing, several future models

KADS identifies itself as a certified Terraplex AG dealer and Terraplex as Revolution's exclusive US distributor. Its current model list says I-19 is shipping, R32 is in final field trials, R37 has a summer

shipping projection, XL40 and XL46 target 2027, and Thunderbird T50/T55 target 2028. An undated "summer" forecast is not proof that shipment occurred. The same page offers paid training and local support and describes component-specific warranty terms, including 1,000 battery cycles and two years for structure and arms. Its broad governmental NDAA claims were not independently substantiated here. KADS sales and model status

The I-19 dealer page says first deliveries occurred in December 2025 and the first 100-plus units came from Brazil, with US manufacturing under development and US software refinement. The tank is 20 gallons fully filled but 19 gallons official load; the dry tank carries 175 pounds. Packages are $37,950 for drone, sprayer, one flight battery/charger, controller without RTK and two controller batteries/charger; $46,250 upgrades to three flight batteries and controller RTK; $51,500 also adds the spreader. Its quoted high daily productivity includes an anecdote flown with the applicator off, which should not be used as a measured spraying trial. KADS I-19 origins, specifications and packages

Revolution's own site quotes an I-19 Kit 1 at $46,500 with drone, controller, charger and three batteries. It groups R32/R37/R40 beside a $72,500 Kit 1 containing drone, controller, lower tank, six batteries and two chargers. The layout does not establish that each named aircraft is available at that price; preserve that ambiguity. Revolution advertised packages The delivered I-19 is a reasonable large-aircraft alternative to evaluate. US ownership, US support and developing domestic production should not be collapsed into an assertion that its delivered Brazilian-origin airframes are American-made throughout.

EXEDY: Japanese parent, Turkish technology lineage, emerging U.S. production

EXEDY's current drone site claims U.S. engineering, manufacturing, support and software from a Michigan operation. Its AYRE CX uses a 15-gallon/56.8-liter liquid tank, 34 Ah battery and centrifugal nozzles. The current product page specifies up to 25 feet of spray width, which remains a condition-dependent claim rather than a validated effective swath for Cuesta. A separate granular module is offered. EXEDY U.S. site AYRE CX specifications

EXEDY's corporate financial report scheduled U.S.-market mass production for September 2026 using technology based on Turkish Baibars products. The Japanese parent, international design lineage, and claimed U.S. manufacturing should all remain visible in an American-options comparison. A scheduled production date is not confirmation of a delivered aircraft. No public USD price or independently verified stocked package was found. Request the current delivery record, actual domestic bill of materials, certifications, battery/charging package and college training terms. The company's product page describes an AcuSpray-powered dealer network in forward-looking language; confirm the servicing dealer rather than assume the network is fully deployed. Corporate production roadmap Dealer and parts proposition

Agricision: Virginia development candidate, still testing

Agricision, Inc. in Virginia describes U.S. design and manufacture using domestic and allied-country components. Its BUBBA-18G is a 70-liter agricultural spray aircraft. Critically, the current homepage also says it is in testing. No public price, complete ready-to-operate package, independently verified delivery, or model-specific certification was established. Its NDAA claim is a manufacturer representation, not a verified Blue/Green listing. Treat this as an emerging demonstration/research lead rather than an immediately procurable teaching system. Its separate Scout imaging/pest-detection proposition should not be conflated with the sprayer. This is Agricision, Inc., not the similarly named British tractor-guidance company. Agricision product and development status

KaiDrones and American Prime: additional emerging watchlist

KaiDrones advertises its Sugar Land-designed/assembled Riso X8 at $25,000 base, with a 2.5-gallon tank and 22-pound payload. Battery/charger contents and loaded aircraft weight are not established. Its homepage uses both launch-year and available-now language, and NDAA-pathway language

alongside stronger compliance assertions. Treat domestic-content, certification and delivered-customer claims as unverified manufacturer representations. The advertised expandable 4-gallon tank does not override the weight limit or establish sub-55-pound eligibility. This is a quote/demo lead rather than a substitute for verified delivery and support. KaiDrones

American Prime explicitly places its multirotor and fixed-wing agricultural aircraft in field validation, with no commercial hardware sales or prices. It offers development partnerships, not a presently purchasable teaching system; production origin and final specifications remain unestablished. American Prime development status

Historical or misclassified entries to remove from the current-manufacturer shortlist

Guardian Agriculture was a genuine US spray-aircraft developer and operator. Its December 2023 company release reported four SC1 aircraft commercially operating in California, 200-pound payloads and up to 60 acres/hour. Guardian commercial operations release It must nevertheless be treated as a historical supplier in a 2026 buying guide. The auctioneer directly documents completed November 2025 Guardian asset auctions, including agricultural aircraft and support equipment. Its former main domain is parked, and no supported new-aircraft sales channel was verified. Old launch prices are not current quotations. Silicon Valley Disposition completed Guardian auction

Rantizo is an agricultural spraying-services and equipment-distribution business, not evidence of a proprietary American airframe. The company announced in August 2025 that its spray operations had been sold to new owners retaining the Rantizo brand while the former parent became American Autonomy, focused on AcreConnect software. Company-issued restructuring announcement Current Rantizo communications describe grower services under new management. It may be an operator or training/service partner, but DJI/XAG resale does not make those aircraft American-made. Rantizo grower update

Imported alternatives and American-label cautions

DJI, XAG and EAVision remain important cost and workflow benchmarks. ABZ provides a Hungarian-made alternative. These should be considered separately because origin, model generation, battery architecture, distribution and regulatory status differ. A U.S. dealer or repair center does not change an aircraft’s manufacturing origin. Exact configured aircraft eligibility still requires the procurement checks below.

DJI Agras

The T25 and T50 provide 20-liter and 40-liter liquid systems respectively; the official manual distinguishes tank volume from maximum operating payload. They are useful comparison points for smaller and medium production work. Newer T25P, T70P and T100 offers exist, but a globally launched aircraft is not proof of U.S. stock, the right battery configuration, or manufacturer warranty. DJI T25/T50 manual

Model and liquid capacity	Advertised equipment price	What is actually included or uncertain	Availability evidence
T25, 20 L	$15,096 kit; $10,999 aircraft/remote	Kit: two DB800 batteries, C8000 station and two 2600 W cables; no generator listed	U.S. dealer listing; confirm exact stock. UAS Agriculture
T50, 40 L	$24,662 kit	Two batteries and charger; adding one listed $2,499 DB1560 makes $27,161 before field power	Estimated 2–4 weeks. Drone Nerds
T25P	$12,399 aircraft/controller	Unavailable variants; dealer kit battery text conflicts with DJI's DB800-family description	Quote and technical reconciliation required. Accurate Ag DJI
T70P, 70 L	$29,199.99 kit	Three batteries, charger, generator and RC Plus 2; exact power variants need confirmation	Sold out. Ares Acres DJI specs
T100, 100 L	$39,999 kit; $21,818 airframe only	Kit: RC Plus 2, spray system/cooler, three DB2160 batteries and C12000 power supply; no generator	Advertised in stock/ready to ship. Drone Nerds

Read charger and warranty terms closely. A second T50 dealer lists a C8000 charger in its $24,519 package, which should be reconciled rather than silently substituted. A different T100 seller lists a $4,999 D14000iEP generator, but its aircraft offer is out of stock and explicitly lacks DJI manufacturer warranty coverage. That seller also specifies sharply different C12000 charging power for three-phase versus single-phase input. These examples are reasons to require an exact, supported electrical bill of materials, not endorsements of a particular combination. T50 alternate package T100 charging and warranty disclosure

XAG

XAG is headquartered in Guangzhou, China. Its own announcement identifies Pegasus Robotics as a U.S. partner, and Pegasus's catalog labels the products XAG. That sales channel is distinct from Hylio's PEGASUS model. Neither a distributor's American location nor the shared name establishes U.S. manufacture. XAG profile XAG/Pegasus partnership Pegasus catalog

Model and liquid capacity	Advertised equipment price	Included equipment and omissions	Status
P100 Pro, 60 L	$19,999 Premium; other advertised packages $17,999–$18,999	Eight B13970S batteries, three CM13600 chargers, parallel accessories, two cooling towers, ARC3 Pro/RTK, XRTK4 rover and tripod; no generator	Listed in stock; 2023-model platform, verify battery age and warranty. Premium kit Catalog
P150, 70 L configuration	$30,000 Spray All Day kit	RevoSpray 4, SRC4, eight batteries, two chargers/cooling towers, XRTK6 Pro and DL1; no generator	Sold out; older generation. Drone Spray Pro
P150 Max, 80 L	$34,830 Standard 14 kW kit	Four B141050 packs, four 3.4 kW supplies, two towers/parallel kits, SRC5 and XRTK7; no engine-driven generator	U.S. sales offer. Standard kit
P150 Max	$17,990 Independence Kit	Sprayer, controller and RTK rover; batteries and chargers absent	Incomplete flight package, not comparable with the Standard total. Independence Kit

XAG announced P150 Max U.S. availability in February 2026. Its single-battery mode reduces the liquid limit to 50 liters; the full 80-liter configuration and advertised fast recharge require the appropriate battery/charger/cooling arrangement. Model generation and usable battery rotations therefore matter

more than simply counting batteries. Local dealer support is advertised, but Cuesta should obtain an actual service location and response commitment. U.S. launch Manufacturer specifications and footnotes

EAVision

EAVision identifies its company as Suzhou Eavision Robotic Technologies in China. The J150 has current U.S. dealer evidence: Agri Spray Drones lists a $40,495 package with controller, three batteries, charger and four-nozzle kit, and a 20-gallon spray tank. AeroAg separately advertises stock. A generator is not listed. The dealer knowledge base identifies 13,000 W fast-charging power, which should be incorporated into a demonstrated power plan. Company identification J150 package AeroAg inventory J150 support specifications

Older J100 prices are not consistent enough for a single dependable figure. Drone Country displays a $19,999 starting figure, $31,995 package text and a stated $1,000 tariff fee. Drone Spray Pro displays $32,995 above conflicting $33,995 description text and sold-out status. Treat those as quote leads, not equivalent delivered-price offers. A separate U.S.-assembled brand is not evidence that an EAVision-branded J150 was manufactured in America. J100 listing Second J100 listing

ABZ Innovation

ABZ says its L-series is engineered and manufactured in Szentendre, Hungary. Its onshoring roadmap and U.S. distribution do not make today's delivered aircraft American-made. The L50 is a meaningful non-Chinese import comparator, but component restrictions, funding eligibility and operating approvals still require exact-model verification. ABZ reported that the FAA agricultural approved-UAS list added L50 on October 2, 2026, alongside L30 V2; inclusion facilitates an operator's process and does not substitute for that operator's own authority. Origin and FCC announcement October FAA-list announcement

The 50-liter L50 ready-to-fly price is $37,132 at Drone Nerds, with a 2–4-week estimate. Another dealer itemizes four smart batteries and a charger/cables but shows sold out. Its 70-liter spreading capability is separate from the spray tank and is not necessarily included. Obtain the actual charger count: ABZ discusses a four-battery rotation using two 6 kW chargers, so a generic description of "a charger" does not establish continuous-operation performance. L50 offer Kit itemization ABZ charging discussion

The L30 is listed at $19,754 without batteries/charger or $34,595 as a full kit at Drone Nerds. PSI advertises lower figures, but its $30,558 full-kit description does not clearly itemize an equivalent flight package. The defensible comparison is a matched quote, not the lowest number. Drone Nerds ABZ catalog PSI L30 kit

The ordinary L10 V2 has a 10-liter tank but a maximum takeoff mass of 29 kg (about 64 lb), so small tank size does not make it a sub-55-pound aircraft. The L10 V2 Light is a separate 5.7-liter, 24.9 kg model marketed specifically for Spain; U.S. availability and eligibility were not verified. L10 V2 Spain-specific Light

Talos, Vector, TopXGun, Yamaha and legacy TTA

Talos T60X is a current U.S.-branded sales option with manufacturing origin unverified in the primary sources reviewed. Talos lists the aircraft/remote bundle at $8,999, in stock, with a 50-liter liquid system expandable to 60 liters. TJ Aerosense lists a $16,999 kit, one in stock, containing three DB2100 batteries, controller, C10000 charger, adapter and cooling station; no generator. Its domestic repair operation is useful, but it does not establish origin. Request the actual manufacturer and supply chain before including Talos in an American-made shortlist. Talos TJ Aerosense kit

Vector AGR/Huida is a Chinese manufacturer with its own published Chinese manufacturing locations. Raptor, formerly HSE-UAV, markets the 70-liter HD580 in the United States. "American-ready" describes market positioning, not manufacturing origin. The manufacturer's newer V-series and a U.S. dealer's HD-series naming should not be assumed interchangeable without a matched configuration. No dependable complete current U.S. price was established in this review. Vector origin Raptor HD580

TopXGun FP700 is a Chinese platform with a 60-liter liquid tank and separate 80-liter spreading system. Its maker lists two 20 Ah batteries in the operating architecture and a 9 kW charger. Quantum Drone Systems markets the aircraft in Michigan, providing a credible U.S. support lead, but no complete published U.S. price was verified there. Unverified marketplace and overseas prices are not used in the budget. Manufacturer U.S. dealer

Yamaha FAZER R is a Japanese unmanned helicopter, not an American-built multirotor. Yamaha currently offers purchase inquiries, identifies up to 32 liters/8.5 U.S. gallons for agricultural payload, and requires its own platform-specific training. The gasoline-powered helicopter avoids the battery-rotation model but introduces a different maintenance and flight-training burden. Price is quote-only in the reviewed source. It is relevant to specialty-crop demonstrations and experienced operators; it is not the recommended first student-owned spray platform. Yamaha FAZER R

TTA/Beijing TT Aviation has historical U.S. distribution and support for M6E-X, G200, M8A-PRO and G300 sprayers. Dealer pages describe 10-, 16-, 20- and 30-liter models, but a current complete USD package and delivery commitment were not established. Retain as a legacy/import research lead rather than an American manufacturer or price leader. TTA company U.S. model descriptions Support library

Custom integrations, service partners and exclusions

Aero Systems West and LahakX are relevant California leads for a custom integration or partner demonstration. ASW publicly offers U.S.-manufactured heavy-lift platforms and spray-payload integration, but it has a Japanese parent, Nippon Kayaku. Its 2024 LahakX partnership announcement concerned development of heavy-payload spraying swarms. LahakX's current site offers spraying as a service or technology licensing from Salinas. Neither source establishes a current, fixed-price retail spraying package. Keep that pathway distinct from buying a complete Hylio, XAG or DJI system. ASW Partnership LahakX

Harris Aerial is a real U.S. heavy-lift manufacturer, but its current catalog emphasizes Carrier platforms and sensor integrations. The identified Stark HX8 spraying announcement is from 2016. No current stocked, priced complete crop-spraying package was verified. A custom solution would need a contemporary spray-system bill of materials, documented calibration, approvals and support. Current Harris catalog Historical spray announcement

Inspired Flight, locally relevant in San Luis Obispo, and **Vision Aerial** are useful imaging/research partners. Their reviewed agricultural material emphasizes optical, thermal, multispectral or other sensing payloads. It does not establish an off-the-shelf agricultural spray package. The same distinction applies to **EagleNXT/AgEagle's eBee** mapping aircraft and **ACSL SOTEN's** multispectral agriculture bundle. Crop monitoring can support a spray program without being its dispensing aircraft. Inspired Flight research Vision Aerial agriculture EagleNXT aircraft ACSL bundle

UAV-IQ BioDrop concerns beneficial-insect/mite release services, a useful integrated-pest-management topic but not a liquid sprayer purchase. Generic heavy-lift, photography, seed-only, firefighting, window-cleaning, and ground-robot products were not counted as agricultural liquid-spray manufacturers without a current complete dispensing proposition. No verified new Watts Innovations spray package was found, so historical PRISM references were also excluded from the purchasing shortlist. UAV-IQ BioDrop Watts product reference

Regulatory and procurement requirements for Cuesta College

Research cut-off: October 15, 2026. This is procurement and curriculum planning guidance, not a legal opinion or authorization to fly or apply pesticides. The college should have its flight-program lead, procurement office, insurer and San Luis Obispo County Agricultural Commissioner confirm the proposed operating model before a purchase order or live application.

1. Classify the operation and fully loaded aircraft first

For agricultural dispensing, FAA's current guidance separates **less than 55 lb at takeoff, including the material carried**, from **55 lb or more**. The former can use Part 107 with applicable agricultural exemptions; the latter uses Parts 91 and 137 with relief from specified Parts 61, 91 and 137 requirements. Exactly 55 lb is outside the small-UAS category. For the small-UAS route, FAA identifies Part 137 exemptions and hazardous-material relief under §107.36; confirm the relief applicable to the actual substance. FAA requires the exemption before the Agricultural Aircraft Operator Certificate (AAOC) application, and requests petitions at least 120 days before the needed effective date. The AAOC applicant and exemption holder names must match. Manufacturer assistance is useful, but neither a purchased aircraft nor a pilot's Part 107 certificate supplies the operator's entire approval package. FAA agricultural dispensing guidance, updated May 5, 2026

Section 44807 is a case-specific FAA safety/exemption route, not a general privilege attached to an American-made drone. Obtain the actual grant, covered make/model/configuration, expiration and conditions; confirm that training, demonstration and operational flights fit its scope. FAA §44807 overview

Water-only does not automatically mean Part 137-exempt. The definition reaches substances intended for plant nourishment, soil treatment, propagation or pest control, as well as dispensing directly affecting agriculture, horticulture or forest preservation. Seeds and fertilizer can therefore be within Part 137 even when they are not pesticides. A pure-water equipment demonstration with no agricultural treatment purpose may be different; that is an operation-specific inference, not a blanket exemption verified here. Describe the exact water demonstration, site, purpose and loaded weight to FAA and obtain a determination before relying on it. 14 CFR §137.3, official CFR text

2. Registration, Remote ID, pilots and teaching status

Register each aircraft through the correct route: FAA's agricultural guidance calls for FA-number/DroneZone registration below 55 lb and N-number/aircraft registration at 55 lb or more. Part 107 registration is per aircraft; a recreational registration is not interchangeable. FAA registration guidance FAA agricultural registration instructions

Drones required to register, or registered voluntarily, generally must comply with Remote ID. Require an exact-model FAA-accepted declaration of compliance and confirm whether the delivered configuration has standard Remote ID or needs a compliant broadcast module. A FRIA is location-specific, not an exemption covering all college flights. FAA Remote ID FAA declaration-of-compliance database

Build the professional curriculum around qualified remote pilots, current knowledge, airspace authorization, visual-line-of-sight operations, maintenance, emergency procedures and the actual exemption/COA conditions. A Part 107 remote pilot must maintain knowledge recency through training every 24 calendar months. FAA pilot certification

Qualifying higher-education organizations can conduct some educational/research flights under §44809, subject to its conditions. **Do not treat that as a blanket educational exemption for a heavy agricultural sprayer or live pesticide work.** FAA's March 2026 education page acknowledges both §44809 and Part 107 pathways. A public college's ownership also does not, by itself, establish that a particular flight is a qualifying public-aircraft operation. The proposed operations should receive a specific regulatory classification. FAA educational users

3. California adds pesticide-specific credentials and supervision

DPR requires a Pest Control Aircraft Pilot Certificate for UAS pest-control work, including private-property applications. Its published unmanned pathway requires appropriate FAA authorization/certification and examinations. The current application packet describes an **apprentice** route, and a **journeyman** route requiring at least one year as an unmanned apprentice plus at least 50 hours applying pesticides with a UAS under a certified journeyman's supervision. Apprentice pest-control operations require direct journeyman supervision. Pilots must register in each county where they work;

continuing education and renewal also apply. These requirements make “complete our short class and become a fully licensed independent spray-drone operator” an inappropriate promise. DPR pilot certification packet, Rev. 7/24

AB 1016 added a separate private-applicator unmanned certificate pathway. However, DPR’s February 2026 enforcement reference states that implementing regulations must first be adopted. The research did not establish that Cuesta has a DPR-accredited program or that a new implementing pathway is currently available. **Obtain written DPR confirmation before promising accreditation, alternative training credit, or eligibility under this pathway.** DPR 2026 citable-sections report California FAC §11902

Pest control **for hire** can also require a DPR Pest Control Business License, a qualified person, financial responsibility and other business requirements. Do not assume that the college’s own-property teaching model and a contractor providing services to growers have the same licensing obligations. DPR business-license packet San Luis Obispo County publishes separate pilot and agricultural-business registration processes. County registration forms

Before any pesticide lab, verify the California-registered product, label directions, lawful aerial application method, minimum carrier volume/rate, drift restrictions, worker/student safety and applicable records. A drone’s ability to spray at a low volume does not override the label. Restricted-material applications add county permitting and notice-of-intent requirements. Fertilizer-only or untreated seed exercises may avoid pesticide-specific rules while still falling under FAA Part 137. DPR labeling guidance DPR product database DPR restricted-material process

4. FCC restrictions: new purchases, existing aircraft and proposals are different

The FCC’s December 22, 2025 action added foreign-produced UAS and critical components, plus the equipment/services described in FY2025 NDAA §1709, to its Covered List. New covered device models cannot receive equipment authorization for U.S. import/sale. The FCC expressly distinguished these from previously authorized models and continued use of lawfully purchased devices. Thus, “all DJI drones are grounded” is not an accurate summary of that action; “a dealer has stock” also does not establish that a particular new purchase is compliant. Verify the exact model, FCC ID, grant status and configuration. FCC fact sheet FCC FAQ

Already effective October 13, 2026: FCC 26-50 prohibits new equipment authorization for devices incorporating certain logic-bearing hardware produced by named Covered List entities where the same entity’s whole device would be prohibited. This is not a ban on every foreign component. The rule is prospective and does not affect previously authorized equipment; pending applications have specified protection unless later amended to change such a component. U.S. assembly alone is consequently insufficient evidence for a new model’s compliance. Official Federal Register final rule, September 11, 2026

Separate proposal: FCC DA 26-758 / PS Docket 26-189 proposes stopping continued import and marketing of specified previously authorized foreign-produced “military-grade” UAS on the Covered List. The proposed categories include aircraft at least 55 lb and aircraft capable of dispensing economic poison, making it directly relevant to agricultural sprayers. Even the proposal expressly preserves use of previously authorized devices already held by users. No final adopting order was located in this research; treat the proposal as procurement/support risk and recheck the docket immediately before contracting. Official Federal Register notice, August 3, 2026

Important current exceptions and support dates:

- FCC exceptions for qualifying Blue UAS-listed equipment and Buy American “domestic end products” extend to **January 1, 2028**. Conditional approvals for covered foreign UAS do not have the old December 31, 2026 sunset if their onshoring and updated-vetting conditions are met. Check the exact approved device, not merely its brand. FCC DA 26-832, page 3, confirming July 21 changes FCC Covered List

- A May 8, 2026 waiver permits qualifying harm-mitigating software/firmware updates for previously authorized covered equipment through at least **January 1, 2029**. This is regulatory permission for qualifying changes, not a guarantee that the manufacturer will support the aircraft. FCC DA 26-454

5. Require separate evidence for each “American” or compliance claim

- **U.S. company / U.S. assembly:** corporate location or final assembly does not prove component origin or supply-chain eligibility. An unqualified “Made in USA” claim has the FTC’s substantially stronger all-or-virtually-all standard. FTC origin guidance
- **NDAA compliance:** request the statutory provisions and exact configured aircraft/components covered by the representation. A sales claim is not evidence of Blue UAS listing, FCC authorization or FAA operating approval.
- **Blue UAS / Green UAS:** require the exact aircraft/configuration and current official listing. DIU distinguishes Blue UAS Cleared from Blue UAS Select; Green UAS is a separate certification path. Verify the current listing rather than assuming Green automatically equals Blue or that a Blue component makes the complete aircraft Blue. DIU program changes DIU updated designations and Green additions
- **Buy American domestic end product:** FAR 25.101 uses U.S. manufacture plus a component-cost test, generally more than 65% domestic content for 2024–2028 deliveries, with exceptions including a COTS component-test waiver and special iron/steel rules. Have procurement determine the applicable test rather than substituting “assembled in America.” FAR 25.101
- **Federal funding / ASDA:** separately screen the FASC/SAM covered-entity list and applicable award terms. The American Security Drone Act’s restrictions on federal funds can reach procurement and operation; FCC eligibility is not a funding-eligibility decision. The statute and FAR provisions should be assessed for Cuesta’s actual funding source, grant or contract, including documented exceptions. ASDA statute, §1825 FAR 52.240-1

Procurement release checklist

1. Record the use cases: basic flight, water-only calibration, seed/fertilizer dispensing, pesticide application, research and off-campus services. Obtain FAA/DPR/county determinations where classification is uncertain.
2. Record empty, battery-equipped and maximum takeoff weights for each configuration; confirm the chosen authorization covers the fully loaded aircraft and training activities.
3. Identify who holds the FAA exemption, AAOC, registration and COA/airspace approvals, and who supplies qualified instructors, remote PICs and any required DPR journeyman supervision.
4. Require model-specific FCC authorization, Remote ID documentation, component-origin disclosure, and separate NDAA/Blue/Green/Buy-American evidence where claimed or required by the funding source.
5. Have procurement approve the funding and district-policy fit; do not rely on a seller’s generic “government approved” statement.
6. Require written spare-parts availability, battery and charger support, software/update commitments, repair turnaround, training scope and protections if delivery or legal marketability changes.
7. Begin with controlled, regulator-cleared training exercises. Make live pesticide labs conditional on all product, site, license, supervision, safety and reporting requirements being satisfied.
8. Recheck FCC orders, exact grants/listings and California curriculum/accreditation status immediately before issuing the purchase order and before the first class.

Verification limit: Several FCC PDFs were available as indexed official FCC text while direct PDF retrieval was blocked. The final-rule and spray-drone-proposal texts were retrieved from the official Government Publishing Office. No direct agency consultation, exhaustive live ECFS docket review, product-specific grant verification or Cuesta-specific legal/funding review was performed.

Budget the complete operating system

Equipment prices are only the first layer

The quote should identify the aircraft, liquid tank, nozzles or atomizers, controller, flight software license, batteries, charger, cooling equipment, and necessary connectors. A granular hopper is a separate capability unless expressly included. Battery counts are not comparable without knowing how many packs the aircraft uses simultaneously: three packs for a single-battery aircraft provide three rotations; three packs for a two-battery aircraft do not.

The second layer is ground infrastructure: a power source with the correct continuous output, voltage, connectors and grounding; water storage; mixing and agitation; transfer pump, filtration and hoses; a controlled landing area; secure transport; and safe battery and chemical storage. A 9 kW charger should not be matched to a generator on the basis of a 9 kW peak rating. Require the supplier to demonstrate the full charging cycle under the proposed electrical configuration. Existing college power, transport, and water infrastructure can reduce this layer substantially, but their suitability has not been verified here.

A useful small-scale price anchor is Quantum Drone Systems' $3,995 Teeny Tiny Tender, a UTV-bed-ready mixing system with a 100-gallon tank, pump and hose reel. It is not a UTV, road trailer, generator, or complete compliant college mixing facility. At the other end, nuWay lists its 2026 Spray Ready trailer at $41,950 and a gooseneck version at $44,999. Those commercial-scale trailers show why a mobile spray business can cost much more than the aircraft; they are not recommended for Cuesta's first teaching installation. QDS compact tender nuWay trailer prices

The third layer is instructional and operating readiness: instructor training and travel, agency applications and recordkeeping, insurance specifically covering the planned activities, site risk controls, maintenance tools and spares, PPE where required, spill response, weather measurement, calibration supplies, and a plan for rinsate and waste. Confirm who teaches, who is pilot in command, who supervises pesticide work, and whether the policy covers students handling equipment. None of those responsibilities comes automatically with an advertised ready-to-fly package.

Illustrative first-year budgets

The following are transparent planning scenarios, not vendor quotations. Aircraft anchors come from the manufacturer and comparison sections. Every other amount is either the identified QDS price or an **author-selected allowance** that must be replaced by a college-specific quote. Existing tow vehicles, building work, salaries, pesticides, land, and large commercial trailer decks are excluded. The tax/freight row is a reserve, not a representation of the applicable California tax rate. No tax exemption has been assumed.

Cost item	Compact domestic teaching: PEGASUS	Domestic mobile application: ARES	Lower-cost import benchmark: XAG P100 Pro
Aircraft equipment anchor	$28,444	$45,999	$19,999
Additional field power/ electrical allowance	$4,000	$6,000	$4,000
QDS compact mixing-system price anchor	$3,995	$3,995	$3,995
Transport, mounting and storage allowance	$3,000	$6,000	$6,000
Instructor training and travel allowance	$2,500	$3,000	$3,000
First-year insurance and licensing/admin allowance	$4,000	$5,000	$5,000
Additional safety, calibration and spares allowance	$1,500	$2,500	$2,500
RTK/corrections allowance	$1,500	$1,500	$1,000
Tax and freight reserve	$4,000	$6,000	$4,000
Subtotal	$52,939	$79,994	$49,494
10% contingency, rounded	$5,294	$7,999	$4,949
Planning total	**$58,233**	**$87,993**	**$54,443**

The XAG package includes electrical chargers, not an engine-driven generator. All three scenarios reserve additional money for suitable power, which must be sized and quoted for the actual charging arrangement. PEGASUS availability remains preorder/delivery-unverified. These prices are not equivalent performance tiers: PEGASUS is a small teaching/spot-spray aircraft, ARES is a larger domestic application platform, and P100 Pro is an imported commercial platform. The table compares the scale of investment, not the price of identical capability. A public grant or district policy may exclude an otherwise cheaper aircraft.

A compact installation using existing suitable campus power and transport could remove some allowances. A commercial raised-deck trailer can add tens of thousands instead. Pyka's aircraft-system price alone is $550,000 during the published current pricing period, so that pathway needs a separately scoped industrial aviation budget and partner justification. The college should not interpret these totals as a ceiling on any project.

Recurring costs and total cost of ownership

Request five-year prices for replacement batteries, propulsion parts, pumps, atomizers/nozzles, hoses and seals, controller repair, scheduled service, freight to the repair center, and software or data subscriptions. Ask for battery warranty conditions, cycle and calendar retirement guidance, storage procedures, and the consequences of a discontinued charger or battery format. Hylio's included flight-software access does not imply every RTK service or third-party application is free. Titan's PEGASUS sheet lists optional DigiFarm corrections at $850 per year, separate from its $1,500 RTK-base option; the proposed workflow must establish which, if either, is needed. PEGASUS price sheet, revision February 23, 2026

For a preliminary worksheet, Cuesta could reserve $10,000 per year after purchase for insurance, batteries, wear parts, software/corrections, recurrent training and field consumables, then replace that round allowance with quotations and measured use. This is a planning choice, not an observed industry average. At that illustrative level, four additional years add $40,000 to the first-year scenarios, before staff time, inflation or major damage. Low flight hours do not eliminate insurance, calendar aging, software charges or staff proficiency requirements.

Battery reserve can be calculated as replacement-set cost divided by the usable planning-cycle count, multiplied by expected annual set cycles, with a separate calendar-life floor. Annual ownership cost should then include that reserve, insurance, service, subscriptions, infrastructure and instructor labor.

Cost per student or completed practical exercise is likely a better college metric than a seller's acres-per-hour or commercial payback claim.

Performance questions that matter for teaching and Central Coast crops

Normalize tank size and throughput

Use U.S. gallons and distinguish liquid volume from allowed payload weight. A dense product can reach the aircraft's weight limit before filling its tank. A hopper volume in liters is not a liquid spray capacity, and maximum lifting capability is not necessarily the approved agricultural payload. The claimed acres per hour may assume a low carrier rate, flat contiguous acreage, rapid ground service and experienced operators. It should not be used as a class scheduling promise.

For scale only, these are arithmetic results for a full tank with no reserve, not product recommendations or legal application rates:

Usable liquid	At 2 gallons/acre	At 5 gallons/acre	At 10 gallons/acre
2.5 gallons	1.25 acres	0.50 acre	0.25 acre
13 gallons	6.50 acres	2.60 acres	1.30 acres
26.4 gallons	13.20 acres	5.28 acres	2.64 acres

Actual treated acreage per sortie is less when a reserve, density limit, partial load, battery limit, turns or transit intervenes. Raising carrier volume from 2 to 10 gallons per acre requires five times as much liquid for the same area. That is why large-row-crop marketing may transfer poorly to a vineyard, orchard, small irregular parcel, or teaching exercise with frequent stops.

Purdue's calibration guidance emphasizes measuring effective swath, rather than accepting maximum advertised width, and checking actual application rate. The acceptance test should measure distribution across overlapping passes at the intended height, speed, carrier rate, crop and weather conditions. Water-sensitive cards, measured refill volumes and exported as-applied data make this a useful student lab. Radar, lidar and terrain following are aids; the operating plan must still identify wires, people, boundaries and other hazards. Purdue spray-drone calibration

Score the whole teaching workflow

For a college, prioritize a predictable instructor override, clear abort and return-to-home behavior, simulator access, readable logs, repeatable calibration, battery-handling ergonomics, repairability, and enough supervised student participation. A single large sprayer does not solve a class of students needing time on the controls. A small aircraft plus structured ground stations for mission planning, loading calculations, calibration, weather, maintenance and documentation may teach more per dollar than the largest tank.

Ask vendors to demonstrate mission import and export with representative Cuesta files. Verify supported coordinate systems and formats, prescription-map execution, offline operation, RTK loss behavior, terrain-model requirements, account access for instructors, and export of as-applied and maintenance records. Do not assume a spraying controller can replace the program's mapping software or that an included FPV camera provides a calibrated crop-imaging workflow.

For local specialty crops, the demonstration should include slope changes, irregular boundaries and relevant canopy geometry. Judge deposition and controllability before speed. A manufacturer reference from similar California work is more useful than a broad claim of global acreage. An instructor-supervised water exercise should be cleared for its actual regulatory status; it is a lower-exposure training method, not a blanket legal exemption.

Suggested implementation and procurement decision

Phase 1: define and demonstrate. Decide whether the immediate goal is an introduction to agricultural application, instructor development, a spray-operator pathway, or a revenue-producing service. Identify a lawful demonstration site, the responsible operator and required supervision. A qualified partner can demonstrate the full workflow while Cuesta develops its own permissions and syllabus. This phase can proceed conceptually without committing to a fleet.

Phase 2: compare a small domestic trainer with one production platform. Obtain identical-scope written quotations for Hylio ARES and the currently supported Central UAS production model. Evaluate PEGASUS separately as a compact option once delivery is confirmed. Add emerging domestic candidates such as EXEDY, Ceres or Revolution after verifying their exact manufacturing origin, current delivery and model-specific eligibility. If policy allows, compare one current DJI or XAG package to quantify the domestic premium and support tradeoffs. Evaluate smaller payloads deliberately, rather than treating lower acreage capacity as an automatic weakness in a school.

Phase 3: buy only after a documented acceptance test and operating plan. Keep the first purchase to one complete system with spares, instructor training and a clear support commitment. Make live-pesticide instruction a separate release gate. Expand to larger aircraft or multiple simultaneous aircraft only when the curriculum, staffing, authorization and demand justify it. Swarming is an advanced operational capability, not an assumed permission or a first-semester requirement.

Questions to put in the request for quotation

1. Identify the legal manufacturer, parent company, exact model, hardware revision, liquid system, controller, radios, batteries and software version. State where design, final assembly, critical components and battery cells originate, and identify what is imported.
2. Provide serial-number stock or a contractual delivery date, deposit and refund terms, warranty scope, loaner/repair arrangements, and at least two comparable U.S. customer references. Separate current deliveries from reservations or prototypes.
3. Itemize batteries, charger, power source, cooling, tank, spray system, controller, RTK, spreader, spares, tools, training, freight and taxes. State every subscription and annual support charge. Confirm whether a listed price is aircraft-only, a configured kit, or a complete operational setup.
4. Supply FAA registration/Remote ID documentation, the actual exemption and approved-model route relevant to the purchase, and exact FCC IDs and authorization evidence. Provide separate NDAA, Green/Blue UAS and domestic-content documentation if applicable. Identify changes caused by the October 2026 rules.
5. Demonstrate aborts, instructor intervention, loss-of-link and GNSS/RTK-loss handling, battery/tank warnings, safe refill procedures, prescription-map operation and exportable logs. Measure deposition and effective swath with the intended teaching configuration.
6. Quote a train-the-instructor program, access for student/simulator use, maintenance training, five-year parts and software plans, and support hours in Pacific Time. State the nearest authorized service location and the actual turnaround commitment.
7. Have Cuesta independently approve insurance, site and chemical controls, FAA/DPR/county requirements, accessibility and student participation, and the permitted funding source. Require any unresolved regulatory or delivery condition to be satisfied before funds become nonrefundable.

Research limits and update triggers

Public sources cannot verify complete supply chains, financial durability, actual stock, or future support. Recheck prices, delivery, exact FCC grants and pending rules, FAA conditions, California licensing, insurance and funding eligibility before contracting and before the first class. Update this report when a model actually ships, support ends, or an agency changes the rules. Resolve contradictory specifications through a written, configuration-specific evidence package.